EXP.NO: 5	Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.

AIM:

To demonstrate the difference between the HTTP GET and POST methods by creating a form in an HTML page and handling the form submissions using two different HTTP methods in a Java Servlet.

ALGORITHM:

1. Input:

- The user fills out a web form with their name and email.
- The form will use both GET and POST methods for data submission.

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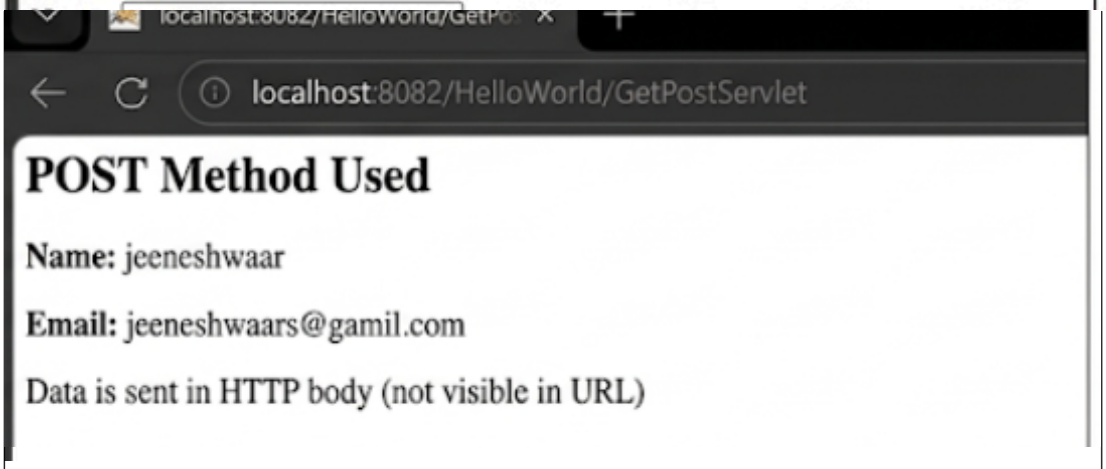
2. Steps for Creating and Handling the HTTP GET and POST Requests:

1. Create an HTML Form:
   - Design an HTMLform with two input fields: one for the name and one for the email.
   - Create two buttons to allow the user to submit the form using GET or POST methods.
   - Use the action attribute to define the Servlet URL and specify the form method (either GET or POST).

2. Create the Servlet to Handle GET and POST Requests:
   - Write a Java Servlet class that will handle bothGETandPOSTrequests.
   - The servlet will have two methods:
     - doGet() to handle the GET requests.
     - doPost() to handle the POST requests.
   - In both methods, retrieve the form data using request.getParameter().
   - In the doGet() method, process and display the data in the URL.
   - In the doPost() method, process and display the data in the HTTP request body.

3. Handle GET Requests in the Servlet:
   - In the doGet() method, use request.getParameter() to retrieve the name and email submitted via GET.
   - Output the data as a response, including the name and email in the URL (visible to the user).
4. Handle POST Requests in the Servlet:
   - In the doPost()method,use request.getParameter() to retrieve the name and email submittedviaPOST.
   - Output the dataasaresponse, displaying the name and email without exposing themintheURL (more secure).
5. Return the Output:
   - The Servlet shouldreturn an HTML page displaying the name and email entered by the user.
   - For the GET method,the output should be visible in the URL.
   - For the POST method,the output should be visible in the response body, not the URL.
6. Deploy and Test the Servlet:
   - Compile the Servlet class and deploy it to a Servlet container (like Apache Tomcat).
   - Test the form using GET and POST methods to see the difference in data submission.
7. Observe and Compare GET and POST Results:
   - Test the GET method: The form data is sentinthe URL.
   - Test the POST method: The form data is sentinthe HTTP request body, not visible in the URL.

Output:



HTML File (form.html):

```
<!DOCTYPE html>
<html> <head>

    <title>GET vs POST Demo</title>
</head>
<body>
    <h2>User Form</h2>

    <!-- GET Form -->
    <form action="GetPostServlet" method="get">
        <h3>Submit using GET</h3>
        Name: <input type="text" name="name"><br><br>
        Email: <input type="email" name="email"><br><br>
        <input type="submit" value="Submit via GET">
    </form>

    <hr>

    <!-- POST Form -->
    <form action="GetPostServlet" method="post">
        <h3>Submit using POST</h3>
        Name: <input type="text" name="name"><br><br>
        Email: <input type="email" name="email"><br><br>
        <input type="submit" value="Submit via POST">
    </form>

</body>
```

```
</html>
```

Java File (GetPostServlet.java)

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;

publicclass GetPostServlet extends HttpServlet {

    //Handle GET request
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String name = request.getParameter("name");
        String email = request.getParameter("email");

        out.println("<html><body>");    out.println("<h2>GET
        Method    Used</h2>");    out.println("<p><b>Name:
        </b> " + name + "</p>"); out.println("<p><b>Email:
        </b> " + email + "</p>"); out.println("<p>Data is
        visible in the URL</p>");    out.println("</body>
        </html>");
    }
```

```
    //Handle POST request
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String name = request.getParameter("name");
        String email = request.getParameter("email");

        out.println("<html><body>");        out.println("<h2>POST        Method
        Used</h2>");  out.println("<p><b>Name:</b>  "  +  name  +  "</p>");
        out.println("<p><b>Email:</b>  "  +  email  +  "</p>");  out.println("
        <p>Data is sent in HTTP body (not visible in URL)</p>"); out.println("
        </body></html>");

    }
}
```

File (web.xml):

```
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee
        https://jakarta.ee/xml/ns/jakartaee/web-app_6_0.xsd"
        version="6.0">
```

```
    <servlet>
        <servlet-name>GetPostServlet</servlet-name>
        <servlet-class>GetPostServlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>GetPostServlet</servlet-name>
        <url-pattern>/GetPostServlet</url-pattern>
    </servlet-mapping>
</web-app>
```

Java Compiled File (GetPostServlet.class) :

```
// Source code is decompiled from a .class file using FernFlower decompiler (from Intellij IDEA).
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.io.PrintWriter;

public class GetPostServlet extends HttpServlet {
    public GetPostServlet() {
    }

    protected void doGet(HttpServletRequest var1, HttpServletResponse var2) throws ServletException, IOException {
        var2.setContentType("text/html");
        PrintWriter var3 = var2.getWriter();
```

```
    String var4 = var1.getParameter("name"); String var5
    =   var1.getParameter("email");   var3.println("<html>
    <body>");          var3.println("<h2>GET          Method
    Used</h2>");   var3.println("<p><b>Name:</b>   "   +
    var4  +  "</p>");  var3.println("<p><b>Email:</b>  "  +
    var5 + "</p>"); var3.println("<p>Data is visible in the
    URL</p>"); var3.println("</body></html>");


  }


  protected void doPost(HttpServletRequest var1, HttpServletResponse var2) throws
ServletException, IOException {
    var2.setContentType("text/html");
    PrintWriter var3 = var2.getWriter();
    String var4 = var1.getParameter("name");
    String var5 = var1.getParameter("email");
    var3.println("<html><body>");
    var3.println("<h2>POST Method Used</h2>");
    var3.println("<p><b>Name:</b> " + var4 + "</p>");
    var3.println("<p><b>Email:</b> " + var5 + "</p>");
    var3.println("<p>Data is sent in HTTP body (not visible in URL)</p>");
    var3.println("</body></html>");
  }
}
```

RESULT:

Thus, A Servlet was created to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.